Cat Image Generation using Generative Models

Deep Learning: Project III Plan

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1 Plan

The aim of the project is to explore and evaluate generative modeling techniques for the task of creating realistic images of cats using the *Cat Dataset* [2]. We will focus on modeling high-dimensional data distributions, optimizing generation quality, analyzing the structure of the learned latent space, and the impact of hyperparameters.

Due to the varying resolutions of the input images, we will implement a preprocessing pipeline to standardize all images to a fixed resolution of 64×64 while maintaining aspect ratios through center-cropping, followed by normalization to $[-1, 1]$. To ensure reproducibility, we use a fixed random seed (42) across all experiments (only one because of computational limitations).

2 Models and Architectures

We will implement and compare the following generative approaches:

- **DCGAN with WGAN-GP loss:** A deep convolutional GAN trained with the Wasserstein gradient penalty ($\lambda_{\text{GP}} = 10$, $n_{\text{critic}} = 5$) for training stability and mode collapse prevention. Latent noise vector $z \in \mathbb{R}^{128}$, batch size 64, trained for 100 epochs.

- **Denoising Diffusion Probabilistic Model (DDPM):** A U-Net-based diffusion model with $T = 1000$ timesteps and a linear noise schedule ($\beta_1 = 10^{-4}$, $\beta_T = 0.02$). Exponential moving average (EMA, decay $= 0.9999$) of model weights is used at inference. Fast sampling via DDIM (50 steps by default). Batch size 16 (limited by VRAM), trained for 200 epochs.

- **VQ-VAE with PixelCNN prior:** A two-phase approach: (1) training a vector-quantized autoencoder (codebook size $K = 512$, embedding dim $= 64$, EMA codebook updates with decay 0.99) and (2) training a 15-layer PixelCNN prior over the discrete latent codes. This allows unconditional sampling through the prior.

3 Hyperparameters and Experiments

We will investigate the impact of the following factors:

1. **DCGAN – Latent Dimension:** We will vary $z_{\text{dim}} \in \{64, 128, 256\}$ (baseline: 128) training for 50 epochs each, measuring FID to assess the effect of latent space capacity on sample quality and interpolation smoothness.

2. **DCGAN – Learning Rate:** We will vary the learning rate for both generator and discriminator among $\{1 \times 10^{-4}, 2 \times 10^{-4}, 4 \times 10^{-4}\}$ (baseline: 2×10^{-4}) while fixing all other hyperparameters.

3. **DDPM – DDIM Inference Steps:** We will evaluate the epoch-200 DDPM checkpoint at $\{10, 25, 50, 100, 200\}$ DDIM steps to characterize the speed–quality trade-off without retraining.

4. **Stability and Mode Collapse:** WGAN-GP loss with one-sided label smoothing (real labels = 0.9) is used for DCGAN. We will monitor training via periodic sample grids and report collapse indicators.

5. **Dataset Complexity:** A DCGAN trained on the combined *Cats and Dogs* dataset [3] (25 000 images) will be compared to the cats-only baseline. FID will be computed separately against cat and dog reference sets.

4 Metrics and Evaluation

Model performance will be assessed using both quantitative and qualitative methods:

- **Quantitative:** We will use the **Fréchet Inception Distance (FID)** to measure the statistical distance between the distributions of real and generated images.
- **Qualitative:** Visual assessment of realism, anatomical correctness (e.g., presence of ears, whiskers), and sample diversity.
- **Latent Space Interpolation:** We will select two generated images and their corresponding latent matrices, then perform linear interpolation to generate 8 additional intermediate steps. We will present the full sequence of 10 images to evaluate the continuity of the latent space.

5 Research Questions

- How does the choice of generative architecture (GAN, Diffusion, VQ-VAE) impact the trade-off between image quality (FID) and computational cost under limited hardware resources (6 GB VRAM)?
- Does the inclusion of dog images in the training set improve the "cat-likeness" of generations through shared animal features, or does it result in hybrid artifacts – reflected as a higher FID against each individual class?
- How does the dimensionality of the DCGAN latent noise vector ($z_{\text{dim}} \in \{64, 128, 256\}$) affect sample quality and the smoothness of latent space interpolation?
- What is the speed–quality trade-off of DDIM inference: at how few steps can the DDPM achieve acceptable FID, and what is lost relative to full reverse-process sampling?

6 Schedule

- **May 19:** Submission of the project plan (KM0).
- **May 20 – June 2:** Data preprocessing (handling multi-resolution images), implementation of selected architectures, and initial training runs.
- **June 3 – June 9:** Hyperparameter tuning, addressing *mode collapse*, FID calculation, and latent space interpolation experiments.
- **June 10 – June 15:** Exploratory extension (Cats vs. Dogs experiment), final result analysis, and preparation of the report and presentation slides.
- **June 16:** Final project deadline and presentation.

7 Bibliography and AI usage

The following AI tools were used in the preparation of this document:

- **GitHub Copilot** (Claude Sonnet 4.6) – used to assist in structuring and refining the content of this plan, including suggesting hyperparameter values, refining research questions, and improving the overall clarity and consistency of the text.
- **Google Gemini** and **DeepL** – used for vocabulary and grammar consultation.

References

References

[1] M. Bujok, S. Kaźmierczak, A. Zajko, A. Zychowski, *Deep Learning project (Course Materials)*, Warsaw University of Technology, 2026.

[2] Weiwei Zhang, Jian Sun, and Xiaoou Tang, Cat Head Detection - How to Effectively Exploit Shape and Texture Features, *Proc. of European Conf. Computer Vision*, vol. 4, pp. 802–816, 2008. Available from `https://www.kaggle.com/datasets/crawford/cat-dataset/data`.

[3] Dogs vs. Cats dataset, Kaggle competition, 2013. Available from `https://www.kaggle.com/competitions/dogs-vs-cats`.